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COURSE CODE: CSA0503

COURSE NAME: Database Management Systems

EXPERIMENT NO: 7 SELECT with various clause – GROUP BY, HAVING, ORDER BY

AIM:

To view the records from the tables using SELECT commands with Group By, Having, Order By

DESCRIPTION:

GROUP BY – HAVING:

➤ The GROUP BY clause groups a set of rows into a set of summary rows by values of columns or

expressions. The GROUP BY clause returns one row for each group. In other words, it reduces the number

of rows in the result set.

➤ The GROUP BY clause is used with aggregate functions such as SUM, AVG, MAX, MIN, and COUNT.

The aggregate function that appears in the SELECT clause provides the information about each group.

➤ The GROUP BY clause is an optional clause of the SELECT statement.

➤ To filter the groups returned by GROUP BY clause, you use a HAVING clause.

ORDER BY:

When you use the SELECT statement to query data from a table, the result set is not sorted in any orders. To sort the

result set, you use the ORDER BY clause. The ORDER BY clause allows you to:

➤ Sort a result set by a single column or multiple columns.

➤ Sort a result set by different columns in ascending or descending order.

ERROR 1054 (42S22): Unknown column 'depart' in 'group statement'
mysql> select Department,count(*) as Students from student group
by department;

Department	Students
CSE	2
IT	2
CSEAI	1
ECE	1

4 rows in set (0.07 sec)

mysql>

mysql> select Department,avg(Score) as Averagescore from student
group by Department;

Department	Averagescore
CSE	84.0000
IT	86.5000
CSEAI	98.0000
ECE	78.0000

4 rows in set (0.07 sec)

SYNTAX:

GROUP BY – HAVING:

SELECT c1, c2,...,cn, aggregate_function(ci)

FROM table

WHERE where_conditions

GROUP BY c1 , c2,...,cn

HAVING conditionS

ORDER BY:

SELECT column1, column2,...

FROM tbl

ORDER BY column1 [ASC|DESC], column2 [ASC|DESC],...

ASC stands for ascending and the DESC stands for descending. By default, the ORDER BY clause sorts the

result set in ascending order if you don't specify ASC or DESC explicitly.

```
mysql> select * from student order by Score asc;
```

RegN0	Name	Gender	Department	Score
1237	Devi	F	CSE	78
1238	Bala	M	ECE	78
1235	Kaviya	F	IT	85
1239	Suresh	M	IT	88
1234	Avanthika	F	CSE	90
1236	Karan	M	CSEAI	98

```
6 rows in set (0.04 sec)
```

```
mysql> select * from student order by Department asc, Name asc;
```

RegN0	Name	Gender	Department	Score
1234	Avanthika	F	CSE	90
1237	Devi	F	CSE	78
1236	Karan	M	CSEAI	98
1238	Bala	M	ECE	78
1235	Kaviya	F	IT	85
1239	Suresh	M	IT	88

```
6 rows in set (0.01 sec)
```

RESULT:

The records from the tables are displayed using SELECT commands with GROUP BY, HAVING and ORDER BY.